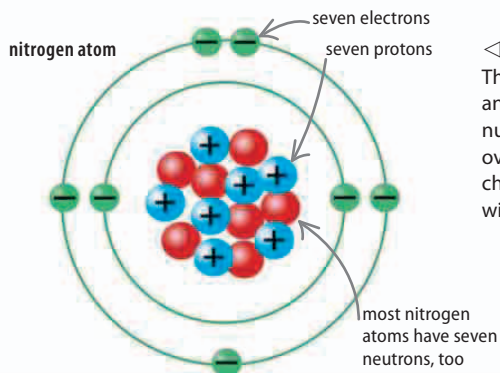


Atoms

Atoms are the building blocks of all material on Earth and out in space (as far as we know). They are not all the same. In fact, every element is made up of its own type of atom. All atoms have positively charged protons in the central nucleus. These are surrounded by negatively charged electrons. The number of electrons and protons varies from element to element and this is what gives each element its properties.



◁ Balanced charge

The number of protons in an atom always equals the number of electrons, so overall the atom has no charge. Neutrons are particles with no, or a neutral, charge.

Reactions

Chemists investigate how elements and compounds behave in reactions. During a chemical reaction, substances known as reactants are transformed into new substances called products. The reaction rearranges the atoms, breaking up the reactants and combining them in new ways to make the products. Most products are different compounds, but some may be pure elements.

sodium (a reactant) combusts when in water (the second reactant)—this reaction produces a liquid (sodium hydroxide) and a gas (hydrogen)

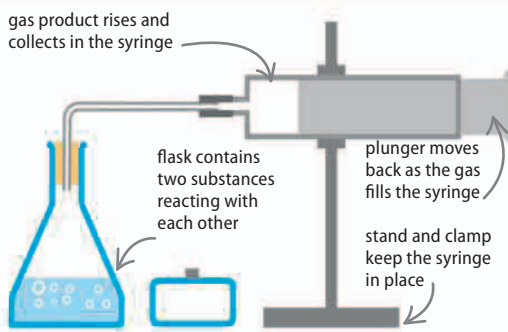


◁ Chemical energy

Reactions take in and release energy, and can be very violent events. Explosions and combustion (burning) are among the most energetic reactions.

Analysis

One role of a chemist is to use knowledge of the physical and chemical properties of different elements and compounds to figure out the content of an unknown substance. This process is called analysis. It involves using a number of tests, such as burning substance (the flame's color gives clues to its contents) or reacting it with a known compound, to see the products created.

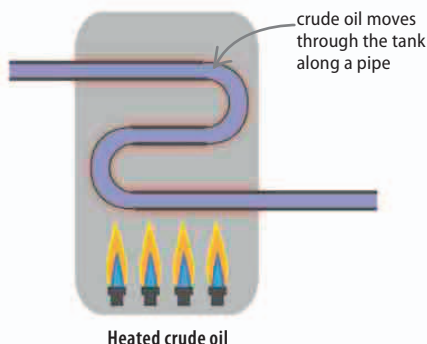


◁ Laboratory apparatus

Chemical reactions are carried out in laboratories. These science workshops contain a range of apparatus for containing and heating reactants, and collecting and measuring products.

Chemical industry

Chemistry is also used to manufacture useful substances. Manufacturing chemicals on an industrial scale is very different to making them in a laboratory. Scientists use their knowledge of what controls the speed of a reaction to come up with the best possible manufacturing process—making the most product for the least expenditure on heat and raw materials.



◁ Petrochemicals

The many hundreds of chemicals in crude oil are used as raw materials for making fuels, plastics, waxes, and medicines. The oil is heated to separate it into different materials (see page 157).